

P2.4.

P1. Solution:

Step 1: Find the general solution of the homogeneous equation $y'' + y = 0$.

It is easy to find

$$y_1(t) = \cos t \text{ and } y_2(t) = \sin t$$

are two linearly independent solutions of the homogeneous equation.

Step 2: Find a particular solution $\chi(t)$ of $y'' + y = \sec t$.

$$W[y_1, y_2](t) = y_1 y_2' - y_1' y_2 = \cos^2 t + \sin^2 t = 1$$

$$\Rightarrow u_1'(t) = -\sec t \sin t = -\tan t$$

$$u_2'(t) = \sec t \cos t = 1$$

$$\Rightarrow u_1(t) = -\int \tan t dt = \ln |\cos t| = \ln \cos t$$

$$\text{since } -\frac{\pi}{2} < t < \frac{\pi}{2}.$$

$$u_2(t) = \int 1 dt = t$$

$$\Rightarrow \chi(t) = u_1 y_1 + u_2 y_2 = \cos t \ln \cos t + t \sin t$$

$$\begin{aligned} \text{Therefore, } y(t) &= C_1 y_1(t) + C_2 y_2(t) + \chi(t) \\ &= C_1 \cos t + C_2 \sin t + \cos t \ln \cos t + t \sin t \\ &= (C_1 + \ln \cos t) \cos t + (C_2 + t) \sin t \end{aligned}$$

P2. Solution:

Step 1. Find two linearly independent solutions of the homogeneous equation $y'' - 4y' + 4y = 0$

Characteristic equation

$$r^2 - 4r + 4 = 0$$

$$(r-2)^2 = 0$$

$$r = 2$$

So that

$$y_1(t) = e^{2t}, y_2(t) = t e^{2t}$$

Step 2. Find a particular solution $\chi(t)$ of

$$y'' - 4y' + 4y = t e^{2t}.$$

$$\begin{aligned} W[y_1, y_2](t) &= e^{2t}(e^{2t} + t e^{2t}) - 2e^{2t} \cdot (t e^{2t}) \\ &= e^{4t} \end{aligned}$$

$$\Rightarrow u_1'(t) = \frac{-g(t)y_2(t)}{W[y_1, y_2](t)} = \frac{-t e^{2t} \cdot t e^{2t}}{e^{4t}} = -t^2$$

$$u_2'(t) = \frac{g(t)y_1(t)}{W[y_1, y_2](t)} = \frac{t e^{2t} \cdot e^{2t}}{e^{4t}} = t$$

$$\Rightarrow u_1(t) = \int -t^2 dt = -\frac{t^3}{3}$$

$$u_2(t) = \int t dt = \frac{t^2}{2}$$

$$\begin{aligned} \Rightarrow \chi(t) &= u_1 y_1 + u_2 y_2 = -\frac{t^3}{3} e^{2t} + \frac{t^2}{2} \cdot t e^{2t} \\ &= \frac{t^3 e^{2t}}{6} \end{aligned}$$

Therefore,

$$\begin{aligned} y(t) &= C_1 y_1(t) + C_2 y_2(t) + \chi(t) \\ &= C_1 e^{2t} + C_2 t e^{2t} + \frac{t^3 e^{2t}}{6} \\ &= e^{2t} \left[C_1 + C_2 t + \frac{t^3}{6} \right] \end{aligned}$$

P3. Solution:

Step 1. Find two linearly independent solutions of $zy'' - 3y' + y = 0$.

$$zy'' - 3y' + y = 0$$

$$r_1 = \frac{1}{2}, r_2 = 1$$

$$\Rightarrow y_1(t) = e^{\frac{1}{2}t}, y_2(t) = e^t$$

Step 2. Find a particular solution $\chi(t)$ of normalized $y'' - \frac{3}{2}y' + \frac{1}{2}y = (t^2 + 1)e^{t/2}$

$$\begin{aligned} W[y_1, y_2](t) &= e^{\frac{1}{2}t} e^t - \frac{1}{2} e^{\frac{1}{2}t} e^t \\ &= \frac{e^{\frac{3}{2}t}}{2} \end{aligned}$$

$$\Rightarrow u_1'(t) = \frac{-\frac{1}{2}(t^2 + 1)e^{\frac{3}{2}t}}{\frac{e^{\frac{3}{2}t}}{2}} = -(t^2 + 1)e^{\frac{1}{2}t}$$

$$u_2'(t) = \frac{\frac{1}{2}(t^2 + 1)e^{\frac{3}{2}t}}{\frac{e^{\frac{3}{2}t}}{2}} = (t^2 + 1)$$

$$\Rightarrow u_1(t) = -\int (t^2 + 1)e^{\frac{1}{2}t} dt$$

$$= -2e^{\frac{1}{2}t}(t^2 - 4t + 9)$$

$$u_2(t) = \int (t^2 + 1) dt = \frac{1}{3}t^3 + t$$

$$\begin{aligned} \Rightarrow \chi(t) &= u_1 y_1 + u_2 y_2 \\ &= -2e^{\frac{1}{2}t}(t^2 - 4t + 9) + e^{\frac{1}{2}t}\left(\frac{1}{3}t^3 + t\right) \\ &= e^{\frac{1}{2}t}\left(\frac{1}{3}t^3 - 2t^2 + 9t - 18\right) \end{aligned}$$

$$\begin{aligned} \Rightarrow y(t) &= C_1 y_1(t) + C_2 y_2(t) + \chi(t) \\ &= C_1 e^{t/2} + C_2 e^t + e^{\frac{1}{2}t}\left[\frac{1}{3}t^3 - 2t^2 + 9t - 18\right] \\ &= C_1 e^{t/2} + \left[C_2 + \frac{1}{3}t^3 - 2t^2 + 9t - 18\right] e^{\frac{1}{2}t} \end{aligned}$$

P5. Solution:

step 1: Find two independent solutions of the homogeneous equation $3y'' + 4y' + 1 = 0$.

$$3r^2 + 4r + 1 = 0$$

$$r_1 = -\frac{1}{3}, r_2 = -1$$

$$\Rightarrow y_1(t) = e^{-\frac{1}{3}t}, y_2(t) = e^{-t}$$

step 2: Find a particular solution $\chi(t)$ of the equation $y'' + \frac{4}{3}y' + \frac{1}{3} = (\sin t)e^{-t/3}$

$$W[y_1, y_2](t) = -\frac{2}{3}e^{-\frac{4}{3}t}$$

$$\Rightarrow u_1'(t) = \frac{-\frac{1}{3}(\sin t)e^{-t}(e^{-t})}{-\frac{2}{3}e^{-\frac{4}{3}t}} = \frac{1}{2}\sin t e^{-\frac{2}{3}t}$$

$$u_2'(t) = \frac{\frac{1}{3}(\sin t)e^{-t}(e^{-\frac{2}{3}t})}{-\frac{2}{3}e^{-\frac{4}{3}t}} = -\frac{1}{2}\sin t$$

$$\Rightarrow u_1(t) = \int \frac{1}{2}\sin t e^{-\frac{2}{3}t} dt = -\frac{3}{26}e^{-\frac{2}{3}t}(3\cos t + 2\sin t)$$

$$u_2(t) = \int -\frac{1}{2}\sin t dt = \frac{1}{2}\cos t$$

$$\Rightarrow \chi(t) = y_1 u_1 + y_2 u_2$$

$$= -\frac{3}{26}e^{-t}(3\cos t + 2\sin t) + \frac{1}{2}e^{-t}\cos t$$

$$= \frac{1}{13}e^{-t}(2\cos t - 3\sin t)$$

$$\text{Thus, } y(t) = c_1 e^{-\frac{1}{3}t} + c_2 e^{-t} + \frac{1}{13}e^{-t}(2\cos t - 3\sin t)$$

$$\text{step 3: } y(0) = 1 \Rightarrow c_1 + c_2 + \frac{2}{13} = 1 \Rightarrow c_1 = \frac{24}{13}$$

$$y'(0) = 0 \Rightarrow -\frac{c_1}{3} - c_2 - \frac{5}{13} = 0 \Rightarrow c_2 = -1$$

$$\Rightarrow y(t) = \frac{24}{13}e^{-\frac{1}{3}t} - e^{-t} + \frac{1}{13}e^{-t}(2\cos t - 3\sin t)$$

P7. Solution:

step 1: Find two independent solutions of the homogeneous equation $y'' - 3y' + 2y = 0$

$$r^2 - 3r + 2 = 0$$

$$r_1 = 1, r_2 = 2$$

$$\Rightarrow y_1(t) = e^t, y_2(t) = e^{2t}$$

step 2: Find a particular solution $\chi(t)$ of the equation $y'' - 3y' + 2y = \sqrt{t+1}$.

$$W[y_1, y_2](t) = 2e^{3t} - e^{3t} = e^{3t}$$

$$\Rightarrow u_1'(t) = \frac{-\sqrt{t+1} \cdot e^{2t}}{e^{3t}} = -e^{-t}\sqrt{t+1}$$

$$u_2'(t) = \frac{\sqrt{t+1} \cdot e^t}{e^{3t}} = e^{-2t}\sqrt{t+1}$$

$$\Rightarrow u_1(t) = \int_0^t -e^{-s}\sqrt{1+s} ds$$

$$u_2(t) = \int_0^t e^{-2s}\sqrt{1+s} ds$$

$$\Rightarrow \chi(t) = y_1 u_1 + y_2 u_2$$

$$= e^t \int_0^t -e^{-s}\sqrt{1+s} ds + e^{2t} \int_0^t e^{-2s}\sqrt{1+s} ds$$

$$= \int_0^t [e^{2(t-s)} - e^{(t-s)}] \sqrt{1+s} ds$$

$$\Rightarrow y(t) = c_1 e^t + c_2 e^{2t} + \int_0^t [e^{2(t-s)} - e^{(t-s)}] \sqrt{1+s} ds$$

$$\text{step 3: } y(0) = 0 \Rightarrow c_1 + c_2 = 0$$

$$y'(0) = 0 \Rightarrow c_1 + 2c_2 = 0 \Rightarrow c_1 = c_2 = 0$$

$$(y(t) = c_1 e^t + c_2 e^{2t} + \int_0^t [2e^{2(t-s)} - e^{(t-s)}] \sqrt{1+s} ds)$$

$$\Rightarrow y(t) = \int_0^t [e^{2(t-s)} - e^{(t-s)}] \sqrt{1+s} ds$$

P9. Solution: $t^2 y'' - 2y = 0$

$$y(t) = t^r \Rightarrow r(r-1)t^r - 2t^r = 0$$

$$\Rightarrow r_1 = -1, r_2 = 2$$

$$\Rightarrow y_1(t) = \frac{1}{t}, y_2(t) = t^2$$

$$y'' - \frac{2}{t^2}y = 1 \quad (\text{from } t^2 y'' - 2y = t^2)$$

$$W[y_1, y_2](t) = \frac{1}{t} \cdot (t^2)' - (\frac{1}{t})' \cdot t^2 = 3$$

$$\Rightarrow u_1'(t) = \frac{-g(t)y_2(t)}{W[y_1, y_2](t)} = \frac{-t^2}{3}$$

$$u_2'(t) = \frac{g(t)y_1(t)}{W[y_1, y_2](t)} = \frac{1}{3t}$$

$$\Rightarrow u_1(t) = \int -\frac{t^2}{3} dt = -\frac{t^3}{9}$$

$$u_2(t) = \int \frac{1}{3t} dt = \frac{1}{3} \ln|t|$$

$$\Rightarrow \chi(t) = u_1 y_1 + u_2 y_2$$

$$= -\frac{t^2}{9} + t^2 \cdot (\frac{1}{3} \ln|t|)$$

$$= t^2(-\frac{1}{9} + \frac{1}{3} \ln|t|)$$

is a particular solution of the equation

$$y'' - \frac{2}{t^2}y = 1$$

Thus,

$$y(t) = c_1 y_1(t) + c_2 y_2(t) + \chi(t)$$

$$= \frac{c_1}{t} + c_2 t^2 + t^2(-\frac{1}{9} + \frac{1}{3} \ln|t|)$$

$$= \frac{c_1}{t} + t^2(c_2 - \frac{1}{9} + \frac{1}{3} \ln|t|)$$